CoCIS Smart Control & Monitoring System - Full Roadmap

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1. Where we started

Coursework: Smart Control & Monitoring System for CoCIS, Makerere University. Requirements: 3 terminals, RFID access control, automated lighting for 2 rooms, SD card storage, serial menus, configurable access schedule with auth, synchronization, Wokwi simulation with ESP-IDF v5.2.1/FreeRTOS in C. Path chosen: ESP-IDF via PlatformIO (ESP-IDF 6.0.1) with Wokwi.

2. 3-Terminal Architecture

Terminal A - Entrance/Access: RFID (MFRC522), entry/exit, validation, in-building count. Terminal B - Room 1: PIR, occupancy, automated lighting with timeout (DONE). Terminal C - Room 2/Monitoring: Room 2 control + central aggregation, logging, menus, SD storage.

3. Room 1 - Done (Working)

platformio.ini: esp32-s3-devkitc-1, espidf. src/main.c: GPIO4 PIR input, GPIO2 LED output, FreeRTOS task room1_task, sample 200ms, 15s timeout, log only on state change. wokwi.toml: points to firmware.bin/elf. diagram.json: PIR on GPIO4, LED+220R on GPIO2. Behavior: Motion -> OCCUPIED LIGHT ON; No motion 15s -> VACANT LIGHT OFF.

4. Room 2 (Next)

Same pattern as Room 1. Create D:\cocis-smart-system\room2 with same files but own project. Same logic - independent first.

5. Entrance/Access Terminal

MFRC522 RFID (SPI), entry/exit logic, track who is inside, publish access events, enforce schedule.

6. Communication Between Terminals

Wokwi allows 1 MCU per project - use 3 separate projects. Connect via MQTT over Wi-Fi. Define event schema (terminal_id, event_type, room_id, value, seq). Need sync to avoid conflicts.

7. Storage, Menus, Schedule, Auth

SD card logging, serial menu for monitoring/config, configurable access schedule, authenticated admin to change settings.

8. Milestones to Deadline (Nov 2)

M0: Room 1 DONE\nM1: Room 2\nM2: Entrance/RFID\nM3: SD + Menus + Schedule + Auth\nM4: MQTT + Sync\nM5: Integration & Testing\nM6: Report, Videos, Submit

9. Next Step

Room 1 is complete and working. Next we will do Room 2 together - you do it yourself with guidance, same pattern as Room 1.